

OMER GADALLA

Aerospace Engineer

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EDUCATION

Middle East Technical University (METU) — (2021–2026)

B.Sc. Aerospace Engineering

EXPERIENCE

Galaxy Rocket Team — (Feb 2025–Present)

● *CFD & Aerodynamics Engineer*

TEKNOFEST Liquid Bipropellant Rocket Competition (A7) | 2026

- Contribute to the design of a **liquid bipropellant rocket** targeting total impulse of 1200 N.s, apogee of 3000 ft and PMF of 0.3 based on the competition requirements.
- Perform **external aerodynamic CFD** to compute drag coefficients and aerodynamic loads, integrating results into structural sizing and flight performance analyses.
- Conduct **multidisciplinary aero-structural analyses** and communicate results to the **system engineering** team to support early design decisions.
- Execute **injector and combustion chamber CFD analyses**, correlating simulation results with **cold flow** test data to assess flow behavior and design robustness.

● *Propulsion Aerodynamics Engineer*

TEKNOFEST Liquid Bipropellant Rocket Engine Development Competition (B3) | 2025-2026

● Design & Analysis Phase

- Contributed to the design, analysis and tests of a **100N liquid bipropellant rocket engine**
- Modeled the **propellant feed system** using MATLAB **Simscape**, analyzing system response and flow behavior under operational conditions.
- Conducted **combustion and CFD analyses** to investigate chamber flow, thermal environment, and performance characteristics.
- Performed **thermo-structural FEA** to assess structural integrity under **high-pressure** and **~3000 K temperature** conditions.

● Testing & Validating Phase

- Assisted with **spray injector cold flow tests**, supporting evaluation of **spray pattern** and **atomization** behavior; and comparing test results with **CFD** simulation data.
- Observed **component-level valve tests**, including **manual and actuated valve functional, leak, and pressure tests**, conducted up to **1.5× MEOP**.
- Achieved **Finalist** ranking among national teams.

MTS Defense Industry — (July 2025–Sep 2025)

R&D Engineer – Intern

- Developed a **conceptual design** for a **20N green bipropellant thruster** for small satellite applications, defining **system architecture** and operational concept.
- Proposed **technical findings and limitations** to stakeholders, highlighting system maturity and development risks.

Technical SKILLS

CAD

- Inventor
- FUSION 360
- SolidWorks

CFD

- Ansys Fluent
- OpenFOAM

Structural Analysis

- Ansys
- Nastran

Programming

- MATLAB
- Simulink
- Simscape
- NASA CEA
- Openrocket

Languages

- Arabic (Native)
- Turkish (Fluent)
- English (Fluent)

CERTIFICATIONS

- MATLAB Onramp
- Simscape Onramp
- Simulink Onramp
- Figes A.Ş. MATLAB & Simulink